Level Set □□ □□□□ □□□

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Level Set Method for Structural Topology Optimization

Single Source of Truth · 2026

> **□ □□□ □□**
>
> □□ "*Implementation of Level Set based Topology Optimization*"□ □□□ □ □□ □□□□(□□□□ □□□·□□□□ □ □□)□ □□□□, □□가 □□□□ □ □□ □□□ □ □□□ □□□□□□□□.
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> □□□ □ □□ □□, □ □□ □□□ □□·□□□ □□□ □□ □□□. 가 □ □□□□ "□□"가 □□□□□□□ □□□ "□□"□ "□□ □□"□□ □□□ □□□.

1. □ Level Set가 — □□ □□

□□ □□ □□□ □□□ "□□□ □□·□□□□ □□ □□□ □□□ □□□ □□□"□ □□□□. □□ □□□□□ □ □ □□ □□□□□□□ □□□□ □□□ □□(boundary)□. □□ □□ □□ □□ □□□ □□ □□ □□ □ □□□□□ □□□, 갸냐 □□□ □□□ □□□□□, □□□ □ □□□ □□□, 가 가 갸냐 가르 □□.

□ "□□□□ □□□□ □□□□ □□"□ □□□□ □□□ □□ □□□□ □□□ Level Set Method□□. □□□□□ □□□□: □□ □□□ □□□□ □□, □□□□ □ □□ □□ □□□□ ψ□ "□□ □□□"□□ □□□ □□□□.

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> □□□ □□ □□□□ □□, □ □□□ □□ □□ □□□□ *ψ(x, t)* □ □□□□. □□□ □□□ □□□□□ □□□ □□ □□□ □□ □□ □□ □□□□□ □□ □ □□.

2. □□□ □□ — □□ □□ □□□ □□

Level Set □□ ψ(x, t)□ □□ □□ □□ "□□□ □□"□□ □□□□ □□. □□□ □□ □□ "□□□(=0)"□ □□□, □□□□ □□□ □□□ □□□ □ □□ ∂Ω가 □□.

- ψ(x) < 0 → □ □□ → □□가 □□ □□ Ω
- ψ(x) > 0 → □ □ → □ □□
- ψ(x) = 0 → □□□ → □□ □□ ∂Ω

□□□ □□□□ □□ □ "□□"□ □□□□□ □□□□ □□ □□ □□. □□ □□□ □□□□□ □□□ □□□□ □□ □□□□ □□□ □□□, 가 □□□ □ □ □□ □□□□. □□□ □ □□□ □□□□ □□ □□□ □, 가 □□ □□□ □□□ □□□ □□□. □□□ Level Set□ □□ □□(topology change)□ □□□□□ □□□□ □□□.

> 가 □□ □□□ □□□ □
> 2D □□ □□ □□□ □□□ 1D □□□□. □ □□□ □□ □□□□□□□□ □□□□□, □□□ □ □□□ □□□□ □ □□□ □□□ □□□ "□□ □□□□ □□□"가 □□□.
>
> Level Set□ □ 1D □□ □□, □□□ □□ □□ 2D □□ ψ□ □□□□. ψ □□□ □□□□□ □□□□ □□ □□□ □□□□□□□□ PDE 갸냐 □□ □□□□□ □□. □□ □□□ ψ□ 0-□□□□ "□□□ □□"□ □□□□ □□□ □□□□.

3. Level Set □□□ SDF

□□□□□ □□□□ □□□□ ψ□ □□ □□□□□□□ □□ □□ Signed Distance Function(□□ □□ □□ □□, SDF)□□ □□□□.

$$\psi(x) = \pm \mathrm{dist}(x, \partial\Omega), \quad |\nabla\psi| = 1$$

SDF □□ — □ □□ □□ "□□□□□ □□ □□ □□"

- ψ□ □□□ □□/□□□ □□□□.
- ψ□ □□□□ □□□□□ □□□ □□□□.

- ψ□ □□ □□ |∇ψ|가 □□□□ 1□□□ □□(Eikonal □□).

□ □ □□□ □□ □□ □□□ □□·□□□ □□□ □□□□□ □□□□ □□□□. □□□ ψ□ □□ □□□□ □□ |∇ψ|=1 □□□ □□□□ □. □□ □□ □-□□□(=□□ □□)□ □□□ □ □ ψ □□□ □□ "□□□□□□□" □□□ □□□ □□ □□ Eikonal □□□□□.

4. Hamilton-Jacobi □□□ — □□□ □□□□ □□□□□

4.1 □□□ □□□ (□□□ □□)

□□ ∂Ω □□ □ □□ □□ □□□□ □□ v(x)□ □□□□□ □□. □ □□ □□ ψ=0 □□ □□□□, □□□ □□□ □ □□□ ψ □□ 0□□□ □□. □□□□ □□□ ψ□ □□ □□□ □□□□ □□□ □□□□□:

$$d\psi/dt = \partial\psi/\partial t + (\nabla\psi)\cdot(dx/dt) = 0$$

□□ □□ □□ ψ=0□ □□□□

□□ □□ ∇ψ/|∇ψ| □□□□ □□ v□□ □□□□ dx/dt = v · ∇ψ/|∇ψ|, □□□□ (∇ψ)·(dx/dt) = v|∇ψ|. □□□ □□□□□ "□□□ □□ □□□ □□"□ □□ □□□ −ωg□ □□ □□ □□ □□□.

$$\partial\psi/\partial t = -v|\nabla\psi| - \omega\, g$$

Hamilton-Jacobi equation (level-set evolution)

4.2 □□ □□

□	□□	□□
−v|∇ψ|	Shape Sensitivity	□□ □□□ □□ □□□□ □□□□(□□□ □□·□□·□□)
−ω g	Topology / Nucleation	□□□ □□□ □□□ □ □□□ □□□(□□ □□)
∂ψ/∂t	□□ □□□	□ □□ □□□ ψ 가 □ □□ □□□□

> □ **|∇ψ|**가 가 □□□
> v□ "□□□ □□ □□"□□, 가 가 □□□ □□□ □□□□ ψ□. □□□ □□ □□□ □ ∇ψ □□□□, □ □□□□ □ □□ □□□ □ ψ가 □□□ □□□□□ |∇ψ|□ □□□□. □□□ □□ v□ □□□□□ □□ □□□□ □□□ □ |∇ψ|가 □□□ □□□ □□□.

5. □□□ — □□□ □□□ □□□□

5.1 Shape Sensitivity v(x)

□□□ □□□□ J(Ω)가 □□□ □□ □□□□ ε□□ □□ □□□ □ □□□ □□□□□ □□□ □□ shape derivative□□, □ 가가 speed v(x)□. □□ □□□ □□□(compliance) 가가 ≡ □ □□□□□ □□□□:

$$v(x) = \sigma(u) : \varepsilon(u) - \lambda$$

□□·□□□ □□□□ □□□□ □□ λ(□□ □□)□ □ □

- v가 □□ "□□□ □□□ □ □□□□" → □ □□□□□.
- v가 □□□ □□□ "□□□ □□" → □ □□□□(□□ □□).

λ□ □□ □□□ □□□ □□ □□□□ □□□, □□□ □□□□ □□□□ 100□□ □□□□ □□(Lagrange Multiplier for volume constraint = 100).

5.2 Topology Sensitivity g(x)

Shape sensitivity□ "□□ □□□ □□□□" □□□ □ □□. 가□ □□□ □□ □□□□ □□□ □□□ □□□□, □ □□□□ □ □□□ v□ □□□ □□□ □□□ □□ □□. □□ □□□□ □□□ topology derivative g(x)□. □□□ □□ 0□ □□ □□□ □ □□ □□□ □ □□□□□ □□□ □□□□□□ □□□□□ □□□ □□□,

- g(x) < 0 → "□□ □□□ □□□ □□□□" → ψ□ □□□□□ □□ □ □□□ □□.
- ω□ topology term□ shape term 가 □ □□□□ □□□ □□, □□ □□□□ □□ □□□□□.

> **□□ □□□**
> □□ □□□□ □□□ **"New holes not created during optimization"** □□□□, □□□□ □□□□□ □□□□ □□ ω=0□□ □□ topology term□ □□□ □□□. □ *shape* □□□□□□ □□□ □□□ □□(□□ □□ level set)□ □□□ □□□□ □□□□□.
>
> □□ Level Set□ □ □□□ □□□□ □□□ □□□□. □□□ level set□ □□ □□□ □ □□□□ □□ □□ □□□. □□ □□□□□ topology derivative, SIMP □□, XFEM □□ □ □□□ □□□ □□□□.

6. Eikonal 가 □□□□ □□□ □□□ — □

Hamilton-Jacobi □□□ □□□□ ψ가 "가 □ □□□□□□ □□ □□ |∇ψ|=1□ □□□. |∇ψ|가 0과 ㄴ □□ □□□□□ □ □□□□ □□ □□□ □□□□, |∇ψ|가 가 □□□ □□□□ □□ □□□ □□□. □□□ □ □□□□ ψ□ "□□ SDF□ □□□□□." □□ □□ PDE가 Eikonal □□□□□.

$$\partial\psi/\partial t + w \cdot \nabla\psi = S(\psi_0), \quad w = S(\psi_0) \cdot \nabla\psi/|\nabla\psi|$$

Eikonal □□□□ (□□□□ 2 □□)

- $S(\psi_0)$□ **smoothed sign function** — □□□ ψ_0□ □□□ □□□□ □□□ □□. □□ □□□□ □□□ □□□.
- w□ □ □□□□ □□□□ □□ □□. "□ □□ □□ □□□□, □□□ □□□ □□" □□□□ |∇ψ|□ 1□ □□□□.
- □□□□□ □□□□ |∇ψ|=1 □ □□□□□ ψ□ □□ □□□ SDF가 □□.

> **□□ □□ □□**
> □□·VFX □□□□□ □□□ □□□ □ □□□ □□□□ Fast Marching Method(FMM), Fast Sweeping Method□ □□ □□. □□□ □□ □□□ □□□□ Eikonal □□□ □□□□□, □□□ 3D(OpenVDB narrow-band □)□□□ FMM□ □□□□.

7. □□ □□ □□□ (□□□□ 2 □□□)

□ □□□□□ □□□ □□ □□□□ □□□□ Word □□□ □□□□□. □□□□ □□ □□ Hamilton-Jacobi·Eikonal □□ §4, §6□□ □□ □□□.

① Mesh the Domain

↓

② Initialize Level Set Function

↓

③ Map Physical Properties with LSF values

↓

④ Solve the Physics (KU = F) for current LSF

↓

⑤ Calculate Shape & Topology Sensitivity

↓

⑥ Evolve the LSF (Hamilton-Jacobi step)

↓

⑦ Converged?

No → ⑧ Re-initialize LSF (Eikonal) → ④□ □□ | Yes → Stop

□ □□□ □□

#	□□	□□□ □□	□□
1	Mesh the Domain	□□ □□ D□ □□□□ □□□ □□□	mesh, □□·□□ □□□
2	Initialize LSF	□□ $\psi_0(x)$ □□(□□ □·□□ □□ □□ □)	□□ SDF ψ_0
3	Map properties with LSF	$\psi \le 0$ □□□ □□ □□, $\psi > 0$□ ersatz(□□ □□)□ E(x) □□	E(x)
4	Solve KU=F	□□ □□ □□□ □□ □□ □□	□□ u, □□ σ, □□□ ε
5	Sensitivity □□	shape v=σ:ε−λ, topology g □□	v(x), g(x)

#	□□	□□□ □□	□□
6	Evolve LSF	Hamilton-Jacobi□ Δt □□ (upwind)	□□□ ψ
7	Convergence check	ΔJ □□ □□·□□ □□ □□ □□	Yes/No
8	Re-initialize	|∇ψ|=1 □□(Eikonal □□ FMM)	□□□ SDF ψ

□□ □□

· ψ□ □□ □□□ 1st-order upwind + TVD-RK(2~3□)□ □□□□ □□□□.

· ψ □□□ CFL □□: Δt ≤ Δx / max|v|. shape sensitivity v가 갸 □□□ v□ □□ □□□ □□ □□□.

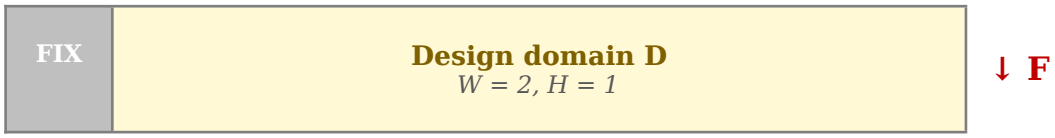
· □□ □□□ □□ λ□ bisection□□ □ □□ □□□□□ augmented Lagrangian□□ □□.

· □□□□□ □ □□ □□ □□ 5~10 □□□□ □ □□□ □□□ □□ □□.

· ersatz □□□ □□□ E_void = 10^{-3}·E_solid □□□ □□ □□ □□□ □□□□ □□.

8. □□ — □□□□ □ (□□□□ 1 □□□)

□□□ □□ □□□□ □□□ □ □□□ □□□□□ □□□□ □□ □□□□. Level Set TO□ "Hello World"□ □□ □□□.



□□ □ □□, □□□ □□□ □□□□ F — □□□□□ $KU = F$

8.1 □□□□

□□	□	□□
Domain width W	2	□□ □□□ □□ □□(□□ □□□)
Domain height H	1	□□ □□
Mesh size	80 × 40	3,200□ Q4 □□. □□ □□ □□ □□□ □□□□□ □□
Volume constraint □□ λ	100	□□□□□ □□ □□□ □□ □□□ □□□ □ □□ □□□□ □□
Iteration	100	□□□ □□ 60~80 □□, □□ □□□□ 100□
ω (topology term)	0 (□□)	□ □□ □□ □□□ → □□ □□ □□□ □□ □□□ □□

8.2 □□□□□□ □□

$$K(\psi)\,U = F, \quad \min_{\psi}\ J(\psi) = \tfrac{1}{2}\,U^{T}K(\psi)U \quad \text{s.t.} \quad \int_{\{\psi\le 0\}} dx \le V^{*}$$

□□ □□ + □□□ □□□ + □□ □□

K(ψ)□ ψ□ □□□ □□ □□□□□ □□ □□. □□□□ J□ □ □□□ □□□(=□□□ □ □□ 2□)□, □□□□ "□□□" □□□. □□ □□□ □□□ □□□ □□ □□□ □□□ — □□ V* □□□□ □□ J□ □□□□□□ □ □□.

8.3 □□□□ □ □□

- ① □□ ψ_0□ □□ □□ □□ □□□□ □□(ω=0□□□ □□□ □□ □□□ □□□ □□□ □□ □□).
- ② Q4 □□□ KU=F□ □□ u, □□□□ σ·ε□ □□□.
- ③ $v(x) = \sigma{:}\varepsilon - \lambda$ □ □ □□ □□□□ □□, □□□□ □□□ □□□□□.
- ④ Hamilton-Jacobi □ □□: $\psi \leftarrow \psi - \Delta t \cdot v|\nabla\psi|$.
- ⑤ 5~10 □□□□ Eikonal □□□□□ $|\nabla\psi|=1$ □□.
- ⑥ ΔJ가 가 □□□ 100□ □□ □ □□.

> 가 □□□□ □
> □□□ □□ □□□□□ "准 가"가 准 □□ □□□ □□. □□ □ □□□ □□ □□□□ □□ □□□ □□□, 가가 □□□ □□□□ □□□ □□□□ □. □□ Michell truss □□□□ □□□□, Level Set TO가 □□□□□□ □□□

> □□.
> □□ ω=0□□□ □□□ □□ □□□ □□□ □□□ □□□□□□, □□□□ □□ □ □□□ □□ □ □□. □□ □□□□□□ topology derivative□ □□□□(ω>0), SIMP□ □□□□□□□ □□ □□□□ □□ □□□ □□□□.

9. □□□ □□

Level Set TO가ㅁ □□□□ □□ □□ □□□□ □□ □□□□. □□ □□ □ □□:

□□	□□□ □□□□□□□	**Level Set□ □□□ □□**
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AM(3D □□□) □□	□□·□□□ □□, □□ □□□	가 □□□ SDF□ □□ STL □□ □ □□□ □□□□ □□
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VFX · □□□□□	OpenVDB narrow-band, □□·□□ □□□	□□□ □□□□□□ □□ □□ □□, FLIP·SPH□ □□ □□
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- □□□□ □□ — Eikonal □□□ □□□□ □□□. → Fast Marching / Fast Sweeping, narrow-band(OpenVDB).
- □□ □□ — v가 □□□□□ ψ가 □□. → v □□□, CFL □□, TVD-RK □□ □□.
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ersatz □□	Ersatz (void) material	□□ □□□ □□ □□ □□□□ □□□ □□ □□□ □□
CFL □□	CFL condition	Δt ≤ Δx/|v|. □□ PDE□ □□ □□ □□
Narrow-band	Narrow-band level set	□□ □□ □ □□ □□/□□□ □□·□□□ □□
Fast Marching Method	FMM	Eikonal□ O(N log N)□□ □□ □□ □□ □□□□
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> □ □□□ §7 □□□□ MATLAB 88-line style□ □ Python/JAX □□□ Level Set TO □□□□□ 가 □□□. □□□ §10□ □□ □ □□(topology derivative □□□, narrow-band, GPU)□ □□□

> 가 □□□.
> VFX □□□ OpenVDB + FLIP fluid □□□□ □□. □□□ Level Set / Eikonal□ □□□ □□□□ □□□ □□□ □□□□ □ □□.